ALAN TSENG

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EDUCATION

M.Sc., Electrical Engineering and Computer Science, National Taiwan University — Aug 2021 - Feb 2024
Overall GPA: 4.12/4.3
Coursework: Blockchain and Big Data, FinTech, Computer Vision, Multimedia Security.

EXPERIENCE

Software/Firmware Engineer — September 2024 -
Server Team, Pegatron — Taipei City, Taiwan

- Developed a Rack Server Testing System for the System Integration Testing Team using Kubernetes, including the provisioning of Kubernetes clusters.

Substitute Military Service — April 2024 - Aug 2024
Department of Conscription Administration, Ministry of the Interior — New Taipei City, Taiwan

- Developed a cross-platform document tracking system for dispatch and receipt records, which can automatically generate printable files, reducing recording time by 30% and avoiding 20% of human errors.
- Developed a RAG-based QA system for the agency's private LLM and established an official LINE account.
- Assisted with hardware and software maintenance for the agency.
- Assisted in recording and editing meetings.

Software Developer Intern — Jul 2022 - Nov 2022
Line Corporation (95M MAUs worldwide, social media company in Asia) — Taipei City, Taiwan

- Increased API development time by 20% and solved the data inconsistency issues across teams by designing a service utilizing Swagger, auto-gen types, and Kubernetes deploy with DNS/LB.
- Led 5 developers to build an NFT merchant management system with NextJS and Spring Boot in Hackathon.
- Gave a talk on Ethereum & dApp development in the company study group.
- Developed a large scalable data-driven B2B System using Nx and collaborating with the global team.

SELECTED PROJECTS AND COURSES

Decentralized Ticketing System - Master's Thesis, Web3 FullStack

- Reduced price fluctuations by 98%, eliminating scalper's profit (100%), by simulating the market, writing 3 ticketing strategies and analysis using SQL.
- Developed a full-stack web3 project that includes a smart contract, website, and mobile app using Next.js, React Native, and Solidity with 95% BDD code coverage.
- Adding Fixed Price ticket purchase feature with Apache Kafka.
- Published in 2023 IEEE International Conference on Metaverse Computing, Networking and Applications.

Rate My Professors TW - Full Stack Project

- Fulfilled scraping rate of 100 KB per second using asyncio, and aiohttp, provided an API using Go and gRPC.
- Achieved 3.8K users visited in 1 year using NextJS with course scheduler, rating system, and PWA feature.
- Developing LLM+RAG, NLP, features to Build Modern GenAI Applications and an Android App.

Drum.io - Course Project

- Won top of the class with 50+ votes by developing a trainable air drum game using TensorFlow.js, Arduino, and WebSockets.

Create your dancing robot - Course Project

- Won top of the class with 52 votes for developing a model control system using C/C++ and OpenGL.

River Cleaning Robot - Hackathon

- Led a team and won a competition with 30+ teams by designing a robot using Raspberry Pi.

Adaptive Driving Beam System - Undergraduate Thesis

- Award a scholarship by the Ministry of Science and Technology, Taiwan, published in the 2020 IPPR Conference on Computer Vision, Graphics, and Image Processing (CVGIP) and Honored for top 1% research & graduation by solving the nighttime vehicle detection and designing an adaptive driving beam using CycleGAN, YOLO, Statistics and SVM.

Certified Developer MLOps Professional by Intel - Course

NLP Course by HuggingFace - Course

Large Language Models with Semantic Search by DeepLearning.ai - Course

Finetuning Large Language Models by DeepLearning.ai - Course

Building Multi Modal Search with Vector Databases by DeepLearning.ai - Course

Docker and Kubernetes: The Complete Guide - Course

Linux: Introduction to Shell Scripting for DevOps - Course

Advanced Commands in Linux - Course

HONORS

- **Research Excellence Award** - Honored for top 1% research & graduation.
- **Intel Andy Grove Scholarship** - Selected with 2000+ applicants worldwide for academic and leadership potential.
- **College Student Research Scholarship** - Awarded by Ministry of Science and Technology, Taiwan.

EXTRA-CURRICULAR ACTIVITIES

- **Lead of Developer Student Club** - Selected with 3000+ global applicants, led a team of 73+ students, and delivered talks on machine learning, PoseNet, Python & Colab, and Flutter.
- **ACM Competitive Programing Club Leader** - Led a group of 20 students in a discussion on programming and practice using UVa Online Judge.
- **Volunteering**
  - Provided technical support at the ART FUTURE exhibition showcasing blockchain applications.
  - Assisted with Developer Student Club (DSC) and Google Developer Groups (GDG) events at SITCON(Student Information Technology Conference) and COSCUP(Conference for Open Source Coders, Users and Promoters).

SKILLS

- Artificial Intelligence(Generative AI, LLM, RAG, NLP, CV, ML/DL, Data Analysis), Web(Frontend/Backend) Development, Cloud-Native Applications, Socket, Chrome Extension, SSG/SSR, MVC, TDD/BDD, AWS, CI/CD, Web Crawling, PWA
- Python, C/C++, JavaScript/TypeScript, Golang, Java
- LangChain, Huggingface, ChatGPT, MLflow, Tensorflow, Scikit-Learn, OpenCV, OpenGL, Numpy, Pandas, Matplotlib, Swagger, FastAPI/Flask, NodeJS, Express, Nx, React(NextJS, React Native), Selenium, Playwright, Mocha, Kafka, Spark, Kubernetes, Grafana, Android, Terraform
- NoSQL(MongoDB), SQL(PostgreSQL, MySQL), Vector(Weaviate)